Dimension, not noise: what limits nuisance leakage in regularised JEPAs

Controlled measurements on data with known factors

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Abstract A JEPA regulariser tells the encoder to spread its embeddings over d dimensions. If the content needs only $k < d$, what fills the rest? We suspected nuisance: lighting, background and pose that differ between views. To test this we need ground truth, so we use 3DShapes, whose six generative factors are known. We split them into content and nuisance, control how often nuisance is resampled between views, and measure leakage as the held-out R^2 of a linear probe. Eight arms (no regulariser, VICReg, SIGReg, sliced and full kernel Stein discrepancy (KSD), MMD, an EMA target encoder, and a rank-k oracle) get matched tuning and 5 held-out seeds, with temporal, CIFAR-10 and autoencoder tiers (747 training runs). All five of our hypotheses about the JEPA arms were refuted. Gaussian regularisers do not leak more than VICReg once each is tuned for content; the regularisation weight decides, and more weight means more leakage for every regulariser. Shrinking d towards k reduces leakage; injected noise never does. A KSD prior matched to a non-Gaussian content law does not help, and plain KSD towards $\mathcal{N}(0, I)$ identifies content best at every law. In the temporal tier, EMA alone leaks more than sliced KSD alone. At the tuned settings, autoencoders leak more than every JEPA arm, the one hypothesis that held.

1 Introduction

A joint-embedding predictive architecture (JEPA) encodes two related inputs and trains a predictor to map one embedding to the other [1], [2]. The objective has a trivial minimum, a constant embedding, so every JEPA needs a second ingredient. One family uses an exponential-moving-average (EMA) target encoder with a stop-gradient, as in I-JEPA, V-JEPA and V-JEPA 2 [1], [2], [3]. Another regularises the batch of embeddings. VICReg constrains variances and covariances [4]. LeJEPA's SIGReg tests one-dimensional projections against a standard Gaussian [5]. KerJEPA uses kernel discrepancies: MMD and kernelised Stein discrepancies (KSD), sliced and unsliced [6]. The Gaussian target has a justification. Klindt et al. show that, within their model class, linear identifiability of JEPA latents holds only for Gaussian latents [7].

That analysis assumes the two views differ only by noise on the shared latent [7], and none of these works has ground-truth nuisance. Real view pairs differ in *nuisance*: lighting, background, pose, augmentation parameters. A distributional regulariser asks the encoder to fill d dimensions with a prescribed spread. If the content occupies fewer than d dimensions, something else must fill the rest, and nuisance is available. Whether this happens, and whether d or an information bottleneck (noise injected into z) controls it, has not been measured, because measuring it requires ground truth for both content and nuisance.

We use 3DShapes [8], a rendered dataset with a full grid of six known factors. We build view pairs with identical content and nuisance resampled independently with probability α. Linear probes on frozen embeddings then give content R^2, nuisance-leakage R^2 and an orthogonal identifiability score, each against ground truth. This is an analysis study, not a methods paper: every regulariser we use already exists.

Contributions. Relative to the 35 works we reviewed (Section 2):

1. A controlled measurement of nuisance leakage for JEPA regularisers (VICReg, SIGReg, sliced and full KSD, MMD, EMA, a rank-k oracle), with ground-truth nuisance whose cross-view dependence α the experimenter sets (Section 4.1, Section 4.4).
2. A joint sweep of the dimensional bottleneck (d below and above k) and an information bottleneck (injected noise σ). On this data, dimension limits leakage and noise does not (Section 4.2).
3. A test of whether KSD with a prior matched to a non-Gaussian Euclidean content law improves identifiability, extending a result reported for MMD on compact manifolds [9]. It does not (Section 4.3).
4. A temporal tier (Section 4.5) and an autoencoder tier comparing reconstruction with prediction (Section 4.6).
5. A robustness slice showing that the regularisation weight λ, more than the target law, decides the leakage comparison.

We fixed five JEPA hypotheses (H1–H5) before the sweep. All are refuted, and we report them as such. A sixth, that autoencoders leak more than JEPAs, is confirmed.

2 Related work

Table 1 condenses the overlap table built from the 35 reviewed papers, each verified against arXiv or the venue. Three points shape our claims.

Kernel regularisers for JEPA are not new. KerJEPA trains JEPAs with sliced and full KSD and MMD, Gaussian and IMQ kernels, and Gaussian and Laplace targets [6]. SPHERE-JEPA uses full MMD and KSD towards the uniform law on the sphere [10]; Rectified LpJEPA uses a rectified generalised Gaussian [11]; VISReg combines variance with a sliced-Wasserstein term [12]; Wang and Isola use a hyperspherical uniformity term for contrastive learning [13]. Our KSD and MMD arms re-implement these. KSD comes from goodness-of-fit testing [14], [15]; the IMQ kernel controls convergence where Gaussian kernels can fail [16]; sliced KSD is due to Gong et al. [17]. Work on Stein operators [18], [19], [20], [21], [22] is the background for our matched and oracle targets, and NP-KSD for implicit models [23] is one of our metrics.

Matched targets and identifiability. Beyond Gaussian Worlds shows that heat-kernel MMD towards a target matched to the latent geometry (sphere, torus) improves linear identifiability on rendered data [9]. Klindt et al. vary the data law under a Gaussian target [7]. Our H3 asks whether this carries over to KSD and to Euclidean non-Gaussian content laws. Without supervision or inductive bias, disentangled factors cannot be identified in general [24].

Nuisance and bottlenecks. The only reviewed works that model view-specific nuisance are contrastive or multi-view information-bottleneck methods. Von Kügelgen et al. prove block-identifiability of content when style is resampled across views [25]; Federici et al. use a multi-view information bottleneck [26]. Arnez et al. argue, in theory only, that SIGReg with fixed isotropic noise in z is an exact information bottleneck [27]. VAEs bottleneck information with a KL weight [28], [29], [30]; WAE matches the aggregate posterior to the prior with MMD [31]; Stein-based VAEs use SVGD [32]. RankMe's effective rank [33] is one of our metrics.

Among the reviewed works, none measures the nuisance leakage of a JEPA regulariser under controlled nuisance, none sweeps d and injected noise together, and none compares EMA with KSD on leakage in a temporal JEPA. These are the cells we claim. We claim no novelty for the regularisers, for matched targets, or for the KSD autoencoder.

Table 1: Condensed overlap table (full version with 35 rows and sources in `notes/related_work.md`). "—" means the work does not address the column.

work	regulariser	target	ground-truth nuisance	d vs k	info. bottleneck	identifiability	objective	data
LeJEPA	SIGReg (sliced)	$\mathcal{N}(0, I)$	—	—	—	informal	JEPA	natural
KerJEPA	KSD, MMD (sliced + full)	Gaussian, Laplace	—	—	—	—	JEPA (EMA)	natural
SPHERE-JEPA	MMD, KSD (full)	uniform on sphere	—	—	—	—	JEPA	natural
Beyond Gaussian Worlds	heat-kernel MMD	matched manifold law	—	partial	—	linear	JEPA	rendered
Klindt et al.	SIGReg	$\mathcal{N}(0, I)$	— (noise only)	open question	—	linear (Gaussian)	JEPA	synth. + rendered
Arnez et al.	SIGReg	$\mathcal{N}(0, I)$	—	—	theory only	assumed	temporal JEPA	none
VICReg	var./cov.	moments	implicit	—	—	—	joint emb.	natural
I-/V-JEPA, V-JEPA 2	EMA + stop-grad	—	—	V-JEPA: implicit (narrow predictor)	—	—	(temporal) JEPA	natural
von Kügelgen et al.	InfoNCE-type	—	yes (style)	$d = k$	—	block	contrastive	synth. + rendered
Federici et al.	MIB	—	yes	—	yes	—	multi-view IB	natural
β-VAE, WAE, InfoVAE	KL, MMD	$\mathcal{N}(0, I)$	—	WAE: discussed	yes (β)	metric (β-VAE)	autoencoder	mixed
this study	VICReg, SIGReg, (sliced) KSD, MMD, EMA, oracle	Gaussian; matched uniform/ Laplace; rank-k	**yes, strength α**	**swept**	**swept**	linear (orthogonal)	JEPA, temporal JEPA, AE	rendered + natural

3 Setup

3.1 Data and views

3DShapes (core). The dataset has 480,000 rendered 64×64 scenes on a full grid of six factors: floor, wall and object hue (10 values each), scale (8), shape (4) and orientation (15) [8]. We downsample to 32×32 and store a memory-mapped `uint8` array. The **content** s is object shape, scale and object hue. The **nuisance** n is floor hue, wall hue and orientation. For probing, hues are encoded as $(\cos, \sin)$ pairs and shape as a 3-dimensional regular-simplex code. The content code is $[\cos h, \sin h, \text{scale}, \text{simplex}(\text{shape})]$, so $k = 6$; the nuisance code has 5 dimensions. A fixed random 10% of the factor grid is held out for probe evaluation and never used in training.

A pair is built by drawing content once, drawing view 1's nuisance uniformly, and resampling each nuisance factor of view 2 independently with probability α (Figure 1). At $\alpha = 0$ the views are the same image; at $\alpha = 1$ their nuisance is independent. Both views receive pixel noise of standard deviation 0.02. Validation confirmed that content is identical across views, and that at $\alpha = 1$ the largest absolute cross-view correlation of the nuisance codes is 0.0099.

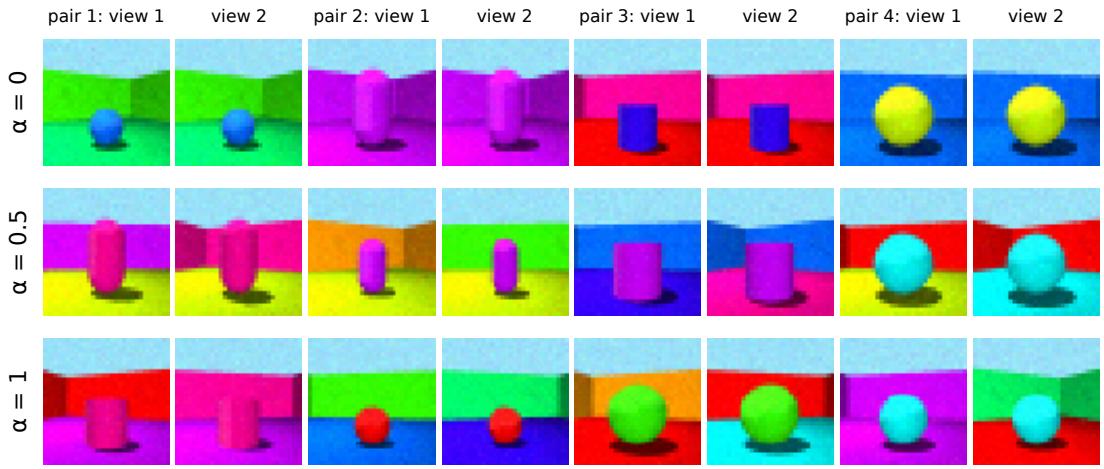


Figure 1: Real 3DShapes view pairs from the pair sampler at three nuisance strengths. Content (object shape, scale, object hue) is shared; floor hue, wall hue and orientation are resampled in view 2 with probability α.

Content law. To test H3, we reweight how often each object-hue and scale value is sampled: uniform (the dataset's grid), or a discretised Gaussian or Laplace law with unit variance on a grid spanning ±2.5 sd. The images stay real; only their sampling frequencies change. Shape stays uniform.

Temporal tier. Clips of $T = 3$ frames: content is fixed, and each nuisance factor takes a ±1 step per frame with probability α (hues wrap; orientation reflects). The predictor sees the concatenated embeddings of frames 1–2 and predicts the embedding of frame 3. We chose clips over Moving MNIST [34] so that this tier shares the core tier's factors and probes.

Natural-image tier. CIFAR-10 [35]: two independent augmentations of one image. The eight augmentation parameters (crop centre, log scale, log aspect, flip, brightness, contrast, saturation) are the nuisance and are independent across views; α scales augmentation strength. Content is probed through the class label on un-augmented test images.

3.2 Objective and bottlenecks

All arms share one encoder f_θ (three 4×4 stride-2 convolutions with 16, 32 and 32 channels, then an MLP $512 \to 256 \to d$, with no normalisation layers so that collapse stays visible) and one MLP predictor g_ϕ ($d \to 256 \to d$). Training uses Adam with learning rate 10^{-3} (10^{-2} for the predictor), batches of 256 pairs and 2000 steps. With $z_v = f_\theta(x_v)$, the image-JEPA loss is

$$\mathcal{L} = \frac{1}{2}\left[\,\mathrm{MSE}(g_\phi(b_\sigma(z_1)), z_2) + \mathrm{MSE}(g_\phi(b_\sigma(z_2)), z_1)\right] + \lambda \cdot \frac{1}{2}\left[\,R(Z_1) + R(Z_2)\right],$$

where R is the arm's regulariser on the batch Z_v. There are two bottlenecks:

- **dimensional:** the embedding dimension $d \in \{3, 6, 12, 24\}$, below and above $k = 6$;
- **information:** $b_\sigma(z) = z + \sigma \cdot \mathrm{sd}_B(z) \odot \xi$, $\xi \sim \mathcal{N}(0, I)$. Noise enters the predictor input, scaled by the detached batch standard deviation, so the per-dimension signal-to-noise ratio is $1/\sigma^2$ whatever the scale of z. Without the scaling, the EMA and unregularised arms could escape the noise by growing z. Probes use the clean z.

3.3 Arms

1. **none**: $R = 0$ (collapse check).
2. **VICReg**: variance hinge plus 0.04 × covariance penalty, the 25:1 ratio of the original paper [4].
3. **SIGReg**: the Epps–Pulley statistic of 256 random one-dimensional projections, computed by the official `lejepa` package [5].
4. **sliced KSD**: for random unit directions g, the one-dimensional IMQ Stein kernel on $t = g^T z$ with projected score $g^T \nabla \log p(z)$ (the $r = g$ variant of sliced KSD [17]), averaged over 32 directions resampled every step.

5. **KSD**: the U-statistic $\widehat{\mathrm{KSD}}^2 = \frac{1}{n(n-1)} \sum_{i \neq j} u_p(z_i, z_j)$ with Langevin–Stein kernel $u_p(x,y) = s(x)^T s(y)\, k(x,y) + s(x)^T \nabla_y k + s(y)^T \nabla_x k + \mathrm{tr}\ \nabla_x \nabla_y k$, score $s = \nabla \log p$ and IMQ base kernel $k(x,y) = (1 + \|x-y\|^2)^{-1/2}$ [14], [16].
6. **MMD**: unbiased MMD² against a fresh prior sample, with Gaussian kernels $\sum_{h \in \{0.5,1,2\}} \exp(-\|x-y\|^2 / 2h^2 d)$.
7. **EMA**: no regulariser. The target is $\mathrm{LayerNorm}(f_\xi(x_2))$ under stop-gradient, with $\xi \leftarrow \tau\xi + (1-\tau)\theta$, as in V-JEPA. Without the target LayerNorm the BYOL-style variant collapsed in the pilot.
8. **oracle** (diagnostic only): KSD towards the rank-k target $\mathcal{N}(0, \mathrm{diag}(1_k, \varepsilon^2 1_{d-k}))$ with $\varepsilon = 0.3$. It uses the known k and is not a practical method.

The target law p is $\mathcal{N}(0, I_d)$ unless stated. For H3, the **matched priors** are product laws with unit-variance coordinates. The Laplace prior has score $-\tanh(x/0.1)/b$, a smoothed sign with $b = 1/\sqrt{2}$. The uniform prior has no defined score, so we use the generalised-normal surrogate with shape 8; MMD and NP-KSD draw exact uniform samples.

The temporal tier runs EMA alone (the V-JEPA recipe), EMA + sliced KSD, EMA + SIGReg, sliced KSD alone and VICReg. The **autoencoder tier** uses the same encoder, a mirrored deconvolutional decoder and per-image squared reconstruction error, plus one of: a β-weighted KL to $\mathcal{N}(0, I)$ (β-VAE [29]); λ·KSD of the deterministic code batch towards the prior (KSD-AE); or λ·MMD (MMD-AE, as WAE-MMD [31]).

3.4 Metrics

Probes are fitted on frozen embeddings of 10,000 training-split images and evaluated on 5,000 held-out test-split images, without view noise and drawn from the run's content law.

- **content R^2**: ridge-probe R^2 of the 6-dimensional content code, averaged over targets (CIFAR: logistic-regression class accuracy);
- **leakage R^2** (primary): ridge-probe R^2 of the 5-dimensional nuisance code (CIFAR: of the 8 augmentation parameters). Lower is better. We report kNN ($k = 10$) R^2 beside it, because the pilot found leakage that only a nonlinear probe detects;
- **identifiability**: R^2 of the best scaled semi-orthogonal map from centred embeddings to the standardised content code. It is 1 for a rotated copy of the content and 0.74 for a random linear mix of it (validation);
- **effective rank**: exp of the entropy of the normalised singular values of the embedding matrix, the quantity RankMe uses [33]. It is meaningless under collapse, so read it with the embedding standard deviation;
- **fit to the prior**: the sliced Epps–Pulley statistic of held-out embeddings, and an NP-KSD statistic [23] of 500 embeddings against the run's target law. We also record the coefficient of variation of the loss over the last 20% of steps.

3.5 Protocol

Matched tuning. Every tuned arm gets three levels, $\lambda \times \{1/5, 1, 5\}$ around a pilot centre (EMA: $\tau \in \{0.99, 0.996, 0.999\}$), on two validation seeds (100, 101). We select the level with the highest mean validation linear content R^2 (CIFAR: class accuracy). Leakage and identifiability are never used, so the selection cannot favour any hypothesis. Core arms are tuned at the **base setting** (3DShapes, $d = 12 = 2k$, $\alpha = 1$, $\sigma = 0$, uniform law), and the d, σ and α slices reuse those levels. H3 arms are re-tuned at the Laplace law; each other tier is tuned at its own base. Table 7 lists the selections. Several land on the lowest level of their grid (Section 6).

Held-out runs and statistics. Each selected configuration is retrained on 5 held-out seeds (1–5), unused in the pilot and in tuning: 495 runs in 99 configurations, none with a non-finite loss. A λ-sensitivity slice reruns all three tuning levels of the six λ-tuned arms at the base on the same seeds (90 runs, 60 of them new). We report means with 95% percentile-bootstrap intervals over seeds (10,000 resamples). Contrasts between arms resample each arm's seeds independently, since arms share no randomness; interactions are differences in differences. With five seeds the percentile bootstrap is somewhat narrow, so we read a contrast as weak if its interval ends within about 0.01 of zero.

Validation gates. Before any sweep, every regulariser was near zero on $\mathcal{N}(0, I)$ samples and clearly larger on collapsed or anisotropic ones (Table 8). SIGReg matched `lejepa` on the same batch and directions (relative difference 0.0), and all losses passed float64 gradient checks. Probes recovered ground-truth factors: content R^2 1.000 from rotated true factors, and leakage R^2 −0.0005 from content alone. The unregularised arm collapsed on real images. SIGReg's statistic is n times the Epps–Pulley statistic and has a positive mean under the null.

4 Results

Table 2 gives every arm at the base setting. The unregularised arm collapses (embedding standard deviation below 0.01 in every seed of every configuration). All regularised arms and EMA learn content. Linear content R^2 stays modest because 3DShapes content is discrete (320 combinations) and no arm is asked to arrange it linearly; kNN content R^2 reaches 0.992 for KSD.

Table 2: Base setting (3DShapes, $d = 12$, $\alpha = 1$, $\sigma = 0$, uniform law): mean over 5 held-out seeds [95% bootstrap CI]. Leakage is the primary metric; "ident." is the orthogonal identifiability R^2.

arm	λ	content R^2	kNN cont.	leakage R^2	kNN leak.	ident.	eff. rank	emb. sd
none	—	0.043 [0.003, 0.086]	0.026	0.080 [0.005, 0.154]	−0.054	0.024 [0.002, 0.050]	6.8	0.000
VICReg	5	0.378 [0.329, 0.407]	0.570	0.673 [0.542, 0.772]	0.683	0.283 [0.219, 0.332]	8.7	1.020
SIGReg	0.05	0.419 [0.370, 0.467]	0.718	0.250 [0.218, 0.281]	0.505	0.268 [0.226, 0.310]	11.0	1.062
sliced KSD	1	0.558 [0.534, 0.578]	0.895	0.019 [0.008, 0.029]	0.130	0.336 [0.307, 0.363]	5.7	1.003
KSD	0.2	0.727 [0.695, 0.748]	0.992	0.002 [−0.001, 0.005]	0.191	0.576 [0.543, 0.595]	10.3	0.817
MMD	10	0.629 [0.602, 0.661]	0.887	0.012 [0.000, 0.026]	0.215	0.354 [0.301, 0.427]	9.9	1.110
EMA	τ = 0.999	0.432 [0.345, 0.526]	0.785	0.215 [0.137, 0.283]	0.347	0.259 [0.196, 0.328]	4.0	0.102
oracle	0.1	0.452 [0.399, 0.501]	0.680	0.451 [0.365, 0.560]	0.577	0.315 [0.228, 0.398]	7.8	0.575

4.1 H1: dimension and isotropic-Gaussian targets

H1: for $d > k$, isotropic-Gaussian regularisers increase nuisance leakage compared with VICReg and EMA. **Refuted under the protocol, and dependent on λ.**

At the tuned λ. At $d = 12$, the Gaussian-target arms leak less than VICReg (0.673 [0.542, 0.772]): SIGReg 0.250 [0.218, 0.281], sliced KSD 0.019 [0.008, 0.029], KSD 0.002 [−0.001, 0.005], MMD 0.012 [0.000, 0.026]. All 8 Gaussian-arm – VICReg contrasts at $d \in \{12, 24\}$ are negative with intervals that exclude zero (8 of 8). Against EMA (0.215 [0.137, 0.283]), sliced KSD, KSD and MMD leak less at $d = 12$, and SIGReg is not different (+0.035 [−0.041, 0.113]). At $d = 24$, SIGReg (−0.057 [−0.169, 0.056]) and KSD (−0.005 [−0.146, 0.134]) are not different from EMA. Among these contrasts, the one result in the direction H1 predicts is nonlinear: SIGReg's kNN leakage exceeds EMA's at every d (at $d = 12$: +0.158 [0.094, 0.217]).

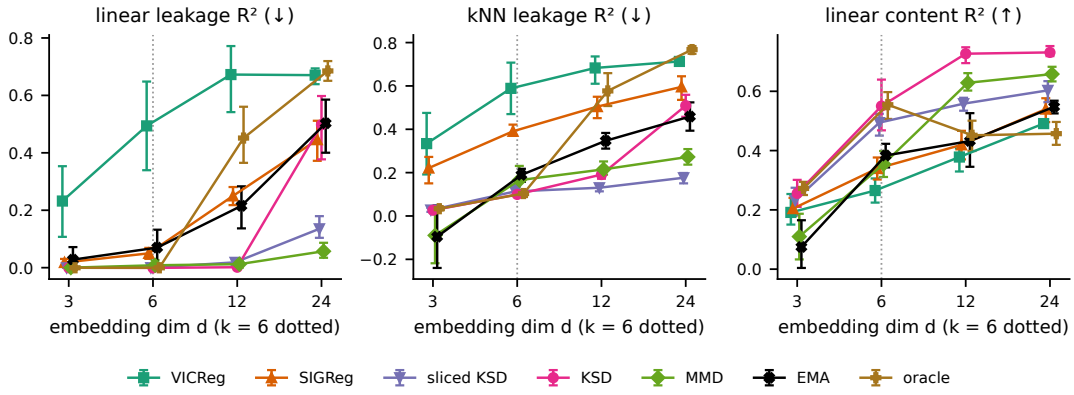


Figure 2: H1: nuisance leakage and content against embedding dimension d (base setting otherwise; tuned λ). Points are means over 5 held-out seeds with 95% bootstrap intervals; the dotted line marks $k = 6$. Every arm leaks more at $d = 24$ than at $d = 3$.

Against d. Every arm, not only the Gaussian ones, leaks more at $d = 24$ than at $d = 3$ (Figure 2), though not monotonically. VICReg is flat from $d = 12$ to $d = 24$; sliced KSD, KSD and MMD stay near zero up to $d = 12$ and then rise. From $d = 3$ to $d = 24$, VICReg goes from 0.232 to 0.671, KSD from 0.000 to 0.498 and EMA from 0.028 to 0.503. Sliced KSD (0.135) and MMD (0.057) stay lowest at $d = 24$. The oracle shrinks the $d - k$ extra directions to standard deviation 0.3 yet still leaks heavily (0.451 at $d = 12$, 0.685 at $d = 24$): linear probes are scale-invariant, so shrinking a direction does not remove the nuisance it carries. At $d = 3$, EMA collapsed in 3/5 seeds and MMD in 2/5, so those two points mix collapsed and non-collapsed runs.

Against λ. We reran all three tuning levels on the held-out seeds (Figure 3). For every regulariser, VICReg included, the highest level leaks more than the lowest, with intervals excluding zero (6 of 6). For KSD the difference is +0.532

[0.458, 0.602]; for VICReg it is +0.499 [0.349, 0.596]. At five times its content-optimal level, each Gaussian-target arm leaks more than VICReg's least-leaky level (0.172 [0.142, 0.206]). For KSD at $\lambda = 1$ the gap is +0.210 [0.146, 0.266], and against EMA +0.166 [0.080, 0.254].

The selection rule (highest validation content R^2) picked the lowest level for sliced KSD and KSD, whose held-out content falls with λ, and the middle level for the other four. Content is not monotone in λ for VICReg (0.265, 0.378, 0.315) or MMD (0.327, 0.629, 0.376), and for VICReg the rule picked a high-leakage level. Against VICReg's least-leaky level, tuned SIGReg leaks more (+0.078 [0.032, 0.120]), in the hypothesised direction, while tuned sliced KSD, KSD and MMD still leak less (e.g. KSD −0.170 [−0.205, −0.140]). The "less than VICReg" part of the verdict therefore rests partly on VICReg's selected level.

Verdict. H1 is refuted for content-tuned regularisers. Under stronger pressure the Gaussian arms do leak more than VICReg and EMA, but more pressure buys leakage for every regulariser, so the effect is not specific to Gaussian targets.

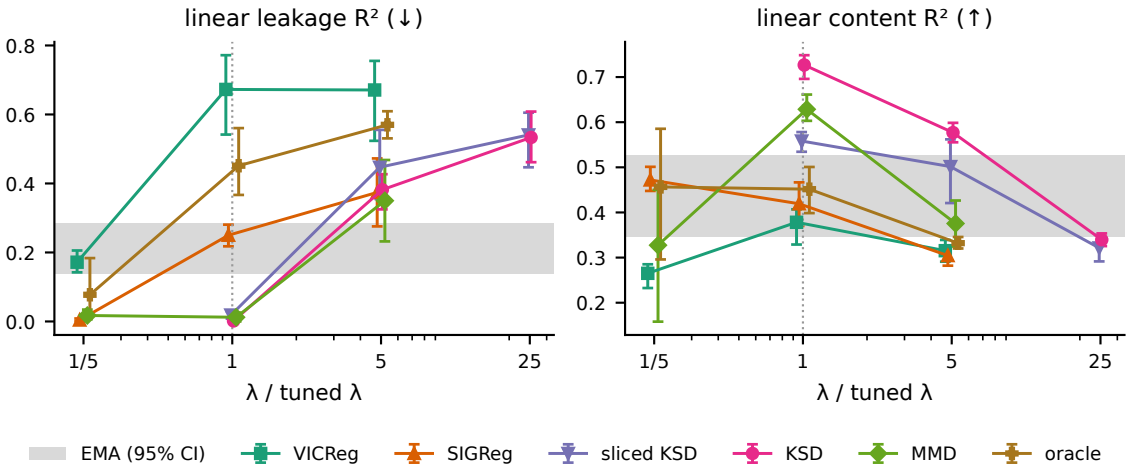


Figure 3: λ sensitivity at the base setting on held-out seeds: leakage and content at each arm's three tuning levels, plotted against λ divided by the tuned λ. The grey band is EMA's 95% interval.

4.2 H2: dimensional against information bottleneck

H2: the information bottleneck (σ) reduces leakage where the dimensional one (d) does not. **Refuted; the opposite holds.**

Narrowing d reduces linear leakage for all six arms ($\sigma = 0$, $d = 6$ against $d = 24$, intervals excluding zero): for example EMA −0.433 [−0.539, −0.322], KSD −0.499 [−0.599, −0.396] and MMD −0.049 [−0.081, −0.020]. For VICReg the evidence is weak (−0.177 [−0.338, −0.012]).

Injected noise never reduces linear leakage (Figure 4). At $\sigma = 1$ against $\sigma = 0$, leakage rises for EMA, SIGReg, sliced KSD, KSD and MMD at both d; for example KSD at $d = 12$ +0.646 [0.596, 0.687] and MMD +0.405 [0.347, 0.469]. VICReg is unchanged (−0.103 [−0.243, 0.050]). Noise also lowers content R^2 for every arm. The only reduction is EMA's kNN leakage (−0.107 [−0.153, −0.060] at $d = 12$), and it comes with a larger loss of kNN content (0.785 → 0.244). The embedding degrades rather than filtering nuisance. EMA's loss is also unstable under noise (coefficient of variation 0.83 at $\sigma = 0.5$, 0.58 at $\sigma = 1$, 0.12 without noise).

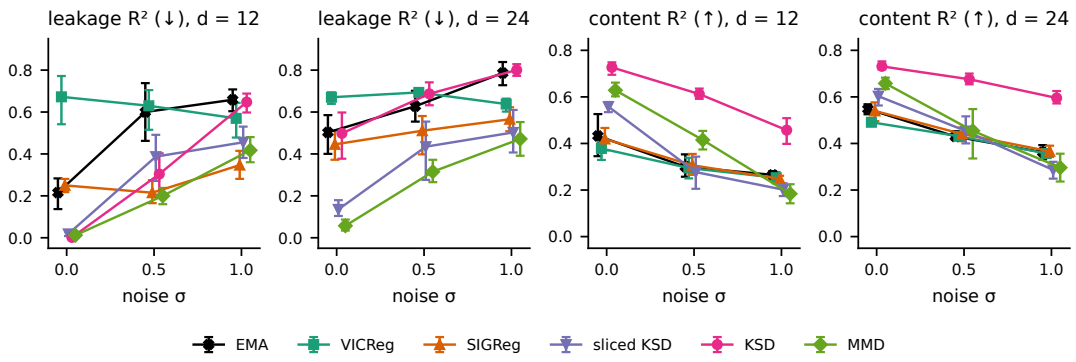


Figure 4: H2: leakage and content against injected noise σ at $d = 12$ and $d = 24$ (α = 1, uniform law, tuned λ). Noise never lowers linear leakage and lowers content for every arm.

4.3 H3: matched priors under non-Gaussian content laws

H3: with non-Gaussian content laws, KSD with a matched prior beats Gaussian-target regularisers on identifiability. **Refuted.**

Uniform law. Matched-uniform KSD (0.554 [0.501, 0.598]) is not different from KSD towards $\mathcal{N}(0, I)$ (0.576 [0.543, 0.595]; difference −0.022 [−0.082, 0.036]). It beats SIGReg, sliced KSD, MMD and the oracle, but so does Gaussian-target KSD. The advantage belongs to full KSD, not to the matched prior.

Laplace law. Matched-Laplace KSD (0.276 [0.235, 0.317]) is far below Gaussian-target KSD (0.772 [0.717, 0.813]; difference −0.496 [−0.558, −0.430]) and also below SIGReg (−0.082 [−0.157, −0.008], weak). Matched sliced KSD is below every Gaussian-target arm at both laws (Figure 5, Table 3).

Across laws. Full KSD towards $\mathcal{N}(0, I)$ has the best identifiability at every law, and its highest value is under the Laplace law; so do SIGReg and sliced KSD, while MMD and the oracle peak under the Gaussian law. Klindt et al. find the opposite for LeJEPA: identifiability peaks at the Gaussian law [7]. The settings differ in ways we cannot separate. Their latents are continuous, d equals the latent dimension, and views differ only by transition noise. Here the law reweights two factors of a discrete grid, $d = 12 > k$, nuisance is present, and probe images are drawn from each law, so scores at different laws come from different populations.

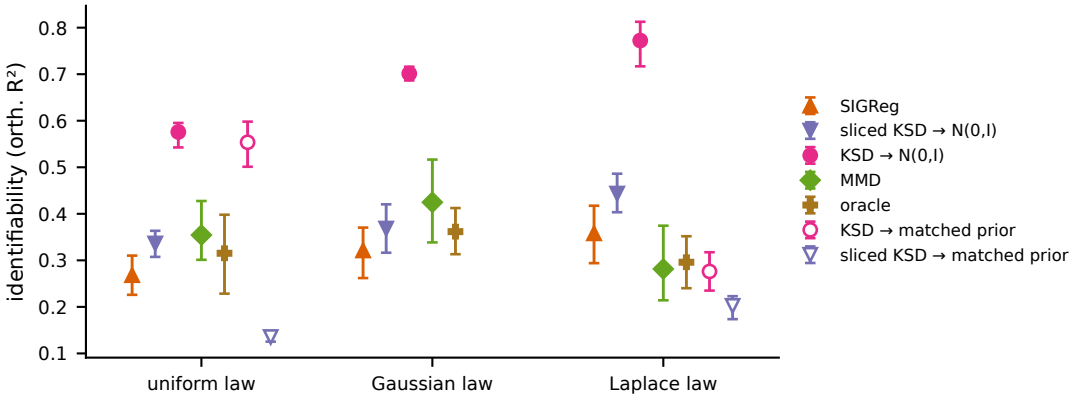


Figure 5: H3: orthogonal identifiability by content law (d = 12). Filled markers: Gaussian targets; open markers: KSD and sliced KSD with a prior matched to the content law (at the Gaussian law the matched prior is $\mathcal{N}(0, I)$).

Three caveats limit this verdict. (1) Laplace-target KSD has little power at batch size 256. In validation it scored −0.0037 on an exact Laplace sample and 0.0040 on a Gaussian one (uniform target: 0.0075 against 3.40). (2) The trained matched-Laplace embeddings do not reach their prior. NP-KSD is 1.14 for KSD and 7.42 for sliced KSD, against 0.02 for Gaussian-target KSD towards its own target. (3) The matched prior acts on all $d = 12$ coordinates, while the law changes only two content factors.

Table 3: H3: identifiability (orthogonal R^2) by content law and target; mean [95% CI] over 5 held-out seeds.

arm (target)	uniform law	Gaussian law	Laplace law
SIGReg	0.268 [0.226, 0.310]	0.321 [0.262, 0.370]	0.358 [0.294, 0.417]
sliced KSD → N(0, I)	0.336 [0.307, 0.363]	0.368 [0.316, 0.420]	0.444 [0.403, 0.486]
KSD → N(0, I)	0.576 [0.543, 0.595]	0.701 [0.687, 0.716]	0.772 [0.717, 0.813]
MMD	0.354 [0.301, 0.427]	0.425 [0.339, 0.517]	0.281 [0.214, 0.374]
oracle	0.315 [0.228, 0.398]	0.362 [0.313, 0.412]	0.296 [0.240, 0.352]
KSD → matched prior	0.554 [0.501, 0.598]	(= N(0, I) row)	0.276 [0.235, 0.317]
sliced KSD → matched prior	0.135 [0.125, 0.145]	(= N(0, I) row)	0.202 [0.174, 0.223]

4.4 H4: EMA under nuisance

H4: EMA/stop-gradient behaves qualitatively differently from distributional regularisers under nuisance. **Refuted for the leakage response; its geometry does differ.**

Raising α from 0 to 1 lowers leakage or leaves it unchanged for every arm, and raises content (Figure 6): EMA −0.458 [−0.537, −0.384], KSD −0.410 [−0.479, −0.332], sliced KSD −0.521 [−0.567, −0.455], MMD −0.336 [−0.376, −0.289], SIGReg −0.113 [−0.173, −0.042], VICReg −0.041 [−0.180, 0.085]. In differences of differences, EMA's response is not different from KSD's (−0.048 [−0.161, 0.055]) or sliced KSD's (+0.063 [−0.036, 0.155]), and EMA's drop is larger than VICReg's (−0.417 [−0.563, −0.260]), SIGReg's (−0.345 [−0.449, −0.246]) and MMD's (−0.122 [−0.214, −0.034]). The size of EMA's response differs from three of the five regularisers but lies inside their range, and no arm responds in the opposite direction (VICReg's interval spans zero). We read this as no *qualitative* difference.

At α = 0.5, EMA leaks 0.548, close to VICReg (0.539); SIGReg leaks 0.338; sliced KSD, KSD and MMD leak 0.006, 0.001 and 0.006. The middle level separates the kernel arms from EMA and VICReg, not EMA from all regularisers.

EMA does differ in geometry. At α = 1 its embedding has standard deviation 0.102 and effective rank 4.0, against 5.7–11.0 for the regularisers.

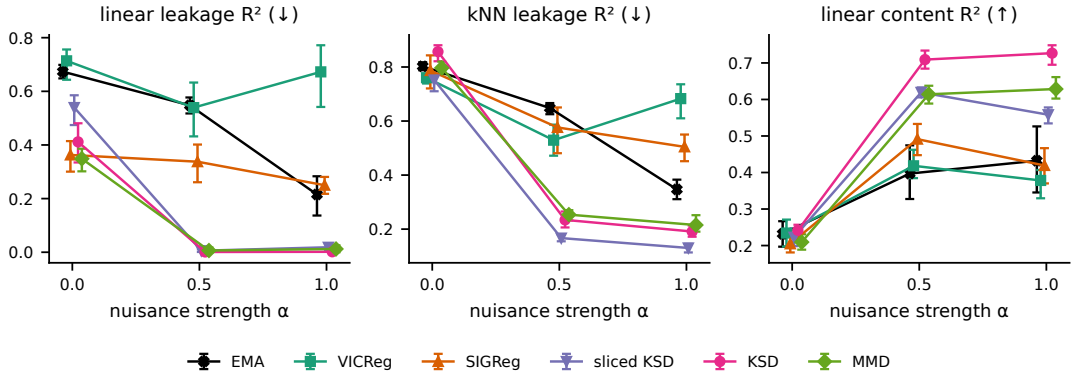


Figure 6: H4: leakage and content against nuisance strength α (d = 12, σ = 0). At α = 0 the two views are the same image, so nuisance is fully predictable from the other view.

4.5 H5: temporal tier

H5: with nuisance that moves over time, EMA + stop-gradient alone leaks less nuisance than any distributional regulariser alone, and adding sliced KSD to EMA leaves leakage unchanged within the CIs. **Refuted on both parts** (Table 4, Figure 7 left).

EMA alone leaks more than sliced KSD alone (+0.385 [0.262, 0.519]) and is not different from VICReg (+0.071 [−0.081, 0.230]; on kNN leakage EMA is higher, +0.212 [0.094, 0.333]). Adding sliced KSD to EMA raises leakage (+0.229 [0.110, 0.337]); adding SIGReg gives +0.121 [−0.005, 0.236]. Leakage is high for every temporal arm except sliced KSD alone (0.098 [0.033, 0.164]). A plausible reason: the target frame's nuisance is one ±1 step from the context, so it is predictable from the context, and an encoder that keeps it lowers the prediction loss.

Table 4: Temporal tier (3DShapes clips, $T = 3$, $\alpha = 1$, $d = 12$). The EMA + X arms keep $\tau = 0.999$ and tune λ.

arm	λ	content R^2	leakage R^2	kNN leakage R^2	eff. rank
EMA	$\tau = 0.99$	0.338 [0.297, 0.384]	0.484 [0.381, 0.601]	0.553 [0.462, 0.644]	7.3
EMA + sliced KSD	5 ($\tau = 0.999$)	0.284 [0.272, 0.296]	0.712 [0.686, 0.741]	0.808 [0.776, 0.844]	12.0
EMA + SIGReg	0.05 ($\tau = 0.999$)	0.251 [0.229, 0.270]	0.605 [0.564, 0.655]	0.779 [0.774, 0.786]	12.0
sliced KSD	1	0.562 [0.510, 0.610]	0.098 [0.033, 0.164]	0.284 [0.232, 0.333]	7.0
VICReg	1	0.393 [0.354, 0.422]	0.413 [0.307, 0.518]	0.341 [0.274, 0.422]	4.3

4.6 H6: autoencoders against JEPA

H6: at matched d, reconstruction-based models leak more nuisance than every JEPA arm. **Confirmed.**

All 21 contrasts (three autoencoders × seven JEPA arms, $d = 12$, linear leakage) are positive with intervals excluding zero (21 of 21), and so are 21 of 21 kNN contrasts (Table 5, Figure 7 right). The smallest gap is β-VAE against VICReg, +0.136 [0.028, 0.267]. The matched-prior question of H3 carries over as a null-to-negative result: under the Laplace law, the Laplace-prior KSD autoencoder has lower identifiability than the Gaussian-prior one (−0.121 [−0.204, −0.036]) and the same content R^2 (−0.011 [−0.081, 0.054]).

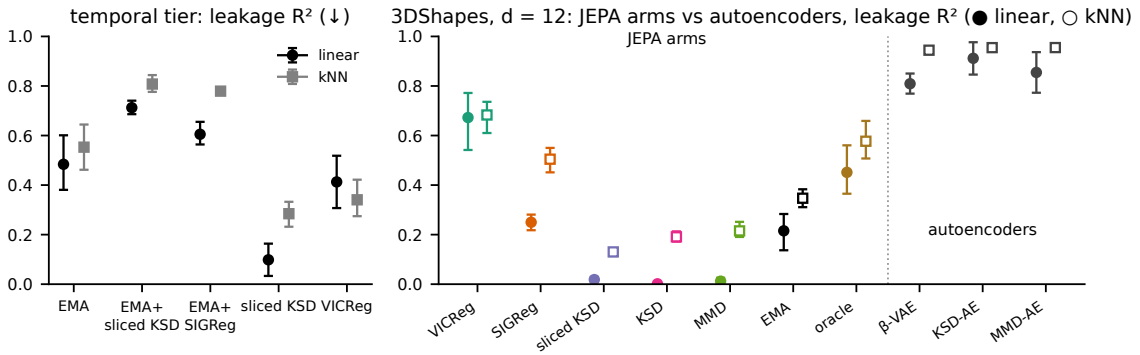


Figure 7: Left: temporal-tier leakage (linear and kNN). Right: leakage of the seven JEPA arms and three autoencoders at $d = 12$ on 3DShapes (filled: linear; open: kNN).

Table 5: Autoencoder tier (3DShapes, $d = 12$); λ / β tuned for content R^2.

model	λ / β	content R^2	leakage R^2	kNN leak.	ident.
β-VAE (KL to N(0, I))	0.2	0.324 [0.292, 0.350]	0.809 [0.769, 0.850]	0.944	0.140 [0.113, 0.168]
KSD-AE ⟶ N(0, I)	10	0.366 [0.313, 0.420]	0.912 [0.846, 0.977]	0.955	0.203 [0.162, 0.243]
MMD-AE ⟶ N(0, I)	20	0.337 [0.270, 0.393]	0.855 [0.773, 0.937]	0.955	0.167 [0.121, 0.208]
KSD-AE ⟶ N(0, I), Laplace law	2	0.485 [0.431, 0.531]	0.957 [0.908, 0.982]	0.966	0.309 [0.263, 0.364]
KSD-AE ⟶ Laplace prior, Laplace law	2	0.474 [0.422, 0.505]	0.935 [0.872, 0.979]	0.963	0.188 [0.123, 0.256]

4.7 Natural images (CIFAR-10)

This tier has no hypothesis (Table 6). Regularised arms reach similar class accuracy, KSD highest (0.357 [0.350, 0.365]) and EMA lowest of the trained arms (0.299 [0.281, 0.316]). The unregularised arm collapses. Linear leakage of the augmentation parameters is small for every arm (0.018 to 0.049, EMA highest), and Gaussian-target arms differ from VICReg by at most a few hundredths. Here the isotropic regularisers do not leak augmentation nuisance linearly to any notable degree. The oracle is omitted because CIFAR has no known k.

Table 6: CIFAR-10 tier ($\alpha = 1$, $d = 12$): class accuracy and linear leakage of the eight augmentation parameters.

arm	λ	class accuracy	augmentation leakage R^2	eff. rank	emb. sd
none	—	0.115 [0.104, 0.128]	−0.021 [−0.061, −0.001]	6.8	0.000
VICReg	5	0.333 [0.324, 0.343]	0.022 [0.014, 0.028]	9.7	1.116
SIGReg	0.05	0.332 [0.324, 0.340]	0.039 [0.031, 0.045]	11.7	0.923
sliced KSD	5	0.348 [0.341, 0.356]	0.036 [0.027, 0.044]	10.6	0.899
KSD	0.2	0.357 [0.350, 0.365]	0.018 [0.012, 0.023]	10.7	0.616
MMD	10	0.341 [0.328, 0.354]	0.020 [0.013, 0.027]	11.2	0.913
EMA	$\tau = 0.999$	0.299 [0.281, 0.316]	0.049 [0.042, 0.057]	4.7	0.073

5 Discussion: dimensional against information bottleneck

The results fit one reading, offered as interpretation, not as a tested claim. A distributional regulariser demands d dimensions of spread. When the shared content supplies fewer ($d > k$), the cheapest way to meet the demand is to spend the spare directions on whatever else varies in the image, here the nuisance. Three manipulations change how hard the demand presses, and for the distributional regularisers all three move leakage as this reading predicts:

- larger d leaves more spare directions (Figure 2);
- larger λ weights the demand more heavily (Figure 3);
- the scaled noise of H2 makes every direction less useful for prediction, which weakens the pull of the prediction loss towards content relative to the regulariser (Figure 4).

Only shrinking d removes room for nuisance. EMA has no explicit demand and its embedding stays low-rank, yet it still leaks once d exceeds k: nothing in a predictive objective pushes nuisance out of the spare directions.

On this reading, the dimensional bottleneck is a filter and the information bottleneck, as we implemented it, is not. Our noise is scaled to each coordinate's batch standard deviation, so the encoder cannot raise the signal-to-noise ratio of content relative to nuisance by rescaling. A bottleneck that charges per bit of $I(z; x)$, such as a learned-variance KL term, or noise fixed in absolute scale as in the analysis of Arnez et al. [27], might behave differently. We did not test those. Also, the predictor reads the noisy z while probes read the clean z, so noise can change what the encoder writes into dimensions the predictor ignores.

The H3 result runs against the motivation for matched targets. Two explanations fit our data, and we cannot separate them: KSD towards a Laplace target has little power at our batch size, so the matched arm gets a weak training signal; and the matched prior constrains all twelve coordinates while only two content factors follow the law. Either way, at this scale the quality of the training signal seems to matter more than the shape of the target.

Finally, the λ-sensitivity slice is the result we would most want a practitioner to see. A study that fixed λ at a conventional value could reach the opposite verdict on H1, and both studies would be correct for their λ. Leakage comparisons between regularisers mean something only together with the rule that set their strengths.

6 Limitations

- **Rendered core data.** 3DShapes has a small discrete factor grid, which bounds content R^2, and its nuisance (colours, orientation) is simpler than real nuisance.
- **Augmentation-defined nuisance.** On CIFAR-10 the nuisance is our augmentation, not a property of the scene, so its leakage measures invariance to our own transformations.
- **Small scale.** A three-layer encoder at 32×32, 2000 steps and $d \leq 24$. All runs trained on one laptop GPU, six at a time (28.4 hours of summed per-run training and evaluation time). Larger encoders and longer training may change every effect.
- **Tuning.** λ was tuned only at the base setting and reused across d, σ and α, so part of the d and σ effects may come from a λ that is no longer optimal there. Several selected levels sit at the lowest end of their grid, so a lower λ might score higher still, and the validation selection is noisy: SIGReg's held-out content R^2 is higher at λ = 0.01 (0.472) than at the selected 0.05 (0.419).
- **Statistics.** Five seeds per configuration, unpaired across arms, with percentile-bootstrap intervals that are somewhat narrow at $n = 5$. The EMA arm is the least stable across seeds.
- **One information bottleneck** (batch-scaled noise on the predictor input) among several possible.
- **Literature coverage.** Novelty claims are relative to 35 works found by a targeted search; 2026 preprints are not exhaustively covered.

7 Future work

Next steps: data with richer nuisance and known factors, such as Causal3DIdent [25] and MPI3D-real; natural images with a held-out nuisance label; real video; an information bottleneck with a learned rate; and λ tuned at every point of the d and σ slices.

8 AI policy

What the AI did. AI agents carried out this study: they designed the experiments within the task specification, wrote the code, ran the experiments, analysed the results and drafted this paper. The agents were Claude models run through Claude Code (Opus 5.5 for the main phases, Sonnet 5.5 for the literature sub-agents; Haiku 5.5 ran only the runner's connectivity check). Sub-agents verified the 35 related-work papers against arXiv or the venue and wrote the BibTeX, and a fresh sub-agent reviewed the finished work. The study ran unattended in five phases; the agent approved each gate itself, with no human review in between. Claude also made the final edit of this text.

What the AI did not do. The task specification (`PROMPT.md`), the runner and the hypotheses were fixed before the sweep. No AI is listed as an author. The author takes responsibility for the paper.

How to read the paper.

- *Numbers* are generated by scripts from per-run results and inserted at render time; none is typed by hand. Rerunning the code checks them.
- *Citations* were verified by AI sub-agents against the source. A human has not re-read each cited paper.
- *Interpretation*, chiefly the Discussion, is the agent's reading. We label it as such and have not tested it.
- *Errors.* Agents can err in ways that tests do not catch. Validation gates, tests and independent review reduce this risk; they do not remove it.

The gate files in the repository record every decision the agents made.

9 Reproducibility statement

All code, configurations and per-run results are in the study repository. `REPRODUCE.md` gives the exact commands, seeds, environment and hardware. Every run is keyed by a hash of its full configuration. The tables in `results/` are generated by `stein.analyze`, and every number, table and figure in this paper is generated from them by `stein.report` and inserted at render time. Seeds are fixed. As a check, we retrained the 25 temporal-tier runs from scratch in an empty run cache on the same machine: 25 of 25 reproduced every metric bitwise. MPS kernels are not guaranteed to be deterministic across hardware or software versions, so reruns elsewhere may differ slightly.

10 Appendix

Table 7: Selected levels at the 3DShapes base setting (validation seeds 100, 101; criterion: mean linear content R^2). Matched-prior, Laplace-law, temporal, CIFAR and autoencoder selections are in `results/tuning_table.md`.

arm (3DShapes base)	selected level	validation content R^2
VICReg	5	0.391
SIGReg	0.05	0.537
sliced KSD	1	0.538
KSD	0.2	0.705
MMD	10	0.610
oracle	0.1	0.438
EMA	τ = 0.999	0.505

Table 8: Validation gate: regulariser value on 256 samples in $d = 12$, mean of 5 draws, for $\mathcal{N}(0, I)$ and four degenerate distributions ("collapsed" = $1 + 0.01\,\mathcal{N}$; anisotropic sd 3/1/0.1; mean shifted by 0.5). VICReg is invariant to a mean shift by design.

regulariser	N(0, I)	collapsed	rank 1	anisotropic	shifted mean
VICReg	0.018	0.986	0.716	0.248	0.020
SIGReg	0.886	278.1	80.8	33.1	31.2
sliced KSD	−0.001	2.096	0.663	0.167	0.159
KSD	−0.001	23.9	4.725	0.365	0.629
MMD	0.000	0.597	0.213	0.060	0.050

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